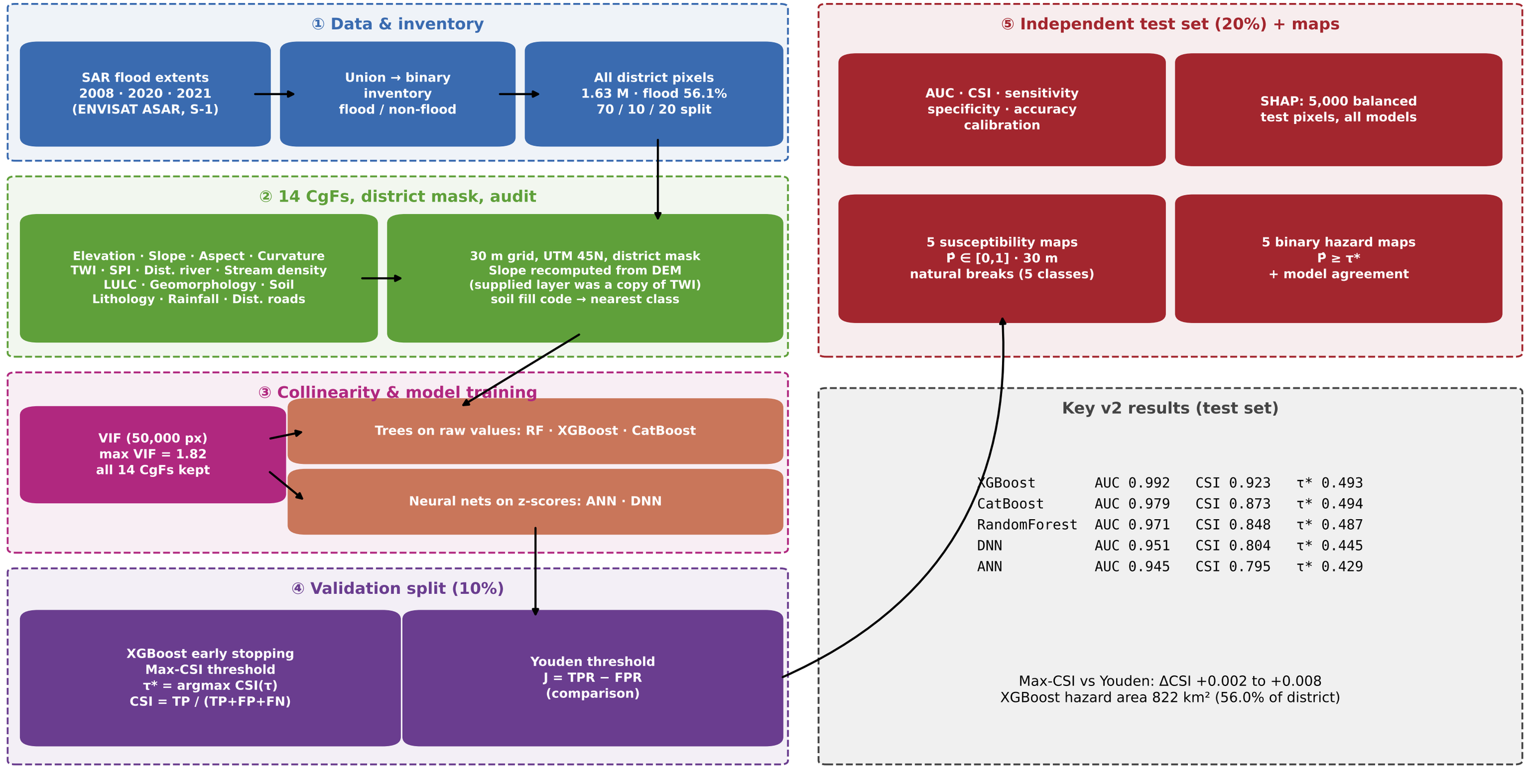

① Data & inventory
SAR flood extents
2008 · 2020 · 2021
(ENVISAT ASAR, S-1)
Union → binary
inventory
flood / non-flood
All district pixels
1.63 M · flood 56.1%
70 / 10 / 20 split
② 14 CgFs, district mask, audit
Elevation · Slope · Aspect · Curvature
TWI · SPI · Dist. river · Stream density
LULC · Geomorphology · Soil
Lithology · Rainfall · Dist. roads
30 m grid, UTM 45N, district mask
Slope recomputed from DEM
(supplied layer was a copy of TWI)
soil fill code → nearest class
③ Collinearity & model training
VIF (50,000 px)
max VIF = 1.82
all 14 CgFs kept
Trees on raw values: RF · XGBoost · CatBoost
Neural nets on z-scores: ANN · DNN
④ Validation split (10%)
XGBoost early stopping
Max-CSI threshold
τ* = argmax CSI(τ)
CSI = TP / (TP+FP+FN)
Youden threshold
J = TPR − FPR
(comparison)
⑤ Independent test set (20%) + maps
AUC · CSI · sensitivity
specificity · accuracy
calibration
SHAP: 5,000 balanced
test pixels, all models
5 susceptibility maps
P ∈ [0,1] · 30 m
natural breaks (5 classes)
5 binary hazard maps
P ≥ τ*
+ model agreement
Key v2 results (test set)
XGBoost AUC 0.992 CSI 0.923 τ* 0.493
CatBoost AUC 0.979 CSI 0.873 τ* 0.494
RandomForest AUC 0.971 CSI 0.848 τ* 0.487
DNN AUC 0.951 CSI 0.804 τ* 0.445
ANN AUC 0.945 CSI 0.795 τ* 0.429
Max-CSI vs Youden: ΔCSI +0.002 to +0.008
XGBoost hazard area 822 km² (56.0% of district)